



Education

- 2023–Present **Master's of Applied Science, Mechanical and Materials Engineering (Co-supervised with Electrical and Computer Engineering)**
Queen's University, Kingston, Canada
4.3/4.3 GPA
- 2018–2023 **Bachelor's of Applied Science, Mechanical Engineering**
With Professional Internship
Queen's University, Kingston, Canada
4.18/4.3 GPA

Employment

- September–Present **Computational Modelling Intern, Canadian Nuclear Laboratories, Chalk River, Canada**
 - Collaborated with experimental scientists to model helium in neutron-irradiated 304 stainless steel from a salvaged component of the National Research Universal reactor.
 - Applied Object Kinetic Monte Carlo simulations for helium and defect distribution and effects.
- Fall 2023 & 2024 **Lead Lab Teaching Assistant, Queen's University, Kingston, Canada**
 - Engineering Graphics Course: Led 2 other TAs in lab sessions of more than 80 students.
 - Live demonstrations, management of sessions and students, tutorials, and marking.
- Winter 2023 **Teaching Assistant, Queen's University, Kingston, Canada**
 - Automatic Control Course: Communicating with students, marking, and proctoring.
- May 2021 – June 2022 **Product Engineering Intern, Hanon Systems Canada, Belleville, Canada**
 - Developed and organized the builds of fluid transport prototypes for automotive thermals.
 - Work included projects for electric vehicles from Ford, GM, and Lucid.
 - Completed 342 prototype build requests under historically high demand.

Research Experience

- 2023–Present **Master's Student Researcher, Queen's University, Kingston, Canada**
Nuclear Materials Group, Computational Materials
 - Development, training, and validation of machine learning interatomic potentials in systems including Na–K–Cl–O, Ni, Si–O, Si–C; involved with Cu–H, Fe–Ni–Cr, Zr–H.
 - Proposed a cost-aware basis set pruning algorithm, optimizing cost-accuracy significantly.
 - Programmed a more optimized CPU and new GPU implementation of the Moment Tensor Potential in LAMMPS.
 - Contributions yielded up to $13 \times$ CPU inference speedups at equal or better accuracy.
- Winter 2023 **Undergraduate Research Project, Queen's University, Kingston, Canada**
Nuclear Materials Group, Computational Materials
 - Development, training, and validation of machine learning interatomic potentials.

Publications

- Internal Review **Meng, Z.**; Zongo, K.; Thoms, M.; Grant, R. E.; Béland, L. K. Accelerating Moment Tensor Potentials through Post-Training Pruning. arXiv preprint arXiv:2510.19737
- Submitted 2025 **Meng, Z.**; Zongo, K.; Torres, E.; Maxwell, C.; Grant, R. E.; Béland, L. K. A Kokkos- Accelerated Moment Tensor Potential Implementation for LAMMPS. arXiv preprint arXiv:2510.00193
- Published 2025 **Meng, Z.**; Sun, H.; Torres, E.; Maxwell, C.; Grant, R. E.; Béland, L. K. Small-cell-based fast active learning of machine learning interatomic potentials. Computational Materials Science 2025, 256, 113919.

Honors and Awards

- 2024 **UNENE Best Student Presentation**
Best student presentation at the UNENE conference in the 7-minute category.
- 2024 **NSERC Alexander Bell CGS-M Scholarship**
The Natural Sciences and Engineering Research Council of Canada's nationwide Master's graduate scholarship.
- 2023 **R. Samuel McLaughlin Fellowship**
- 2023 **L. M. Arkley Prize**
Awarded for the best fourth-year Mechanical Engineering paper and presentation.
- 2023 **Colin T. Bayne Memorial Award**
Awarded to the graduating Mechanical Engineering student who, in the opinion of the Department, has shown most proficiency in innovative design.
- 2023 **Conn-Gilbert Award**
Awarded to the graduating Mechanical Engineering student with the highest average in core Thermodynamics courses.
- 2023 **3rd Place—Professional Engineers Ontario, Kingston, Engineering Competition**
- Various **Dean's Scholar**
- 2022 **Lorne C Elder Scholarship**
- 2021 **Best Project and Best Game Award, CuHacking Hackathon**
- 2018 **Lena MacNeil Scholarship**

Presentations and Posters

- Presentation 2024 **Simulating Complex Chemistry for Advanced Reactors: Machine Learning Potentials in Action**
UNENE Student Presentations, 7-minute category
- Poster 2023 **Systematic Small-Cell Training Set Selection**
UNENE Student Posters